**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a map of a building, and your task is to count the number of its rooms. The size of the map is $n \times m$ squares, and each square is either floor or wall. You can walk left, right, up, and down through the floor squares.

### Input

The first input line has two integers $n$ and $m$: the height and width of the map.

Then there are $n$ lines of $m$ characters describing the map. Each character is either `.` (floor) or `#` (wall).

### Output

Print one integer: the number of rooms.

### Constraints

- $1 \leq n, m \leq 1000$

### Example

Input:
```
5 8
########
#..#...#
####.#.#
#..#...#
########
```

Output:
```
3
```